Fiscal Year:	FY 2025	**Task Last Updated:**	FY 01/31/2025
PI Name:	Lundblad, Nathan Ph.D.		
Project Title:	Microgravity Dynamics of Bubble-Geometry Bose-Einstein Condensates		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	04240-6018	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013 Fundamental Physics NNH13ZTT002N (Cold Atom Laboratory--CAL)
Start Date:	04/01/2014	**End Date:**	09/30/2025
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 9/30/2025 per D. Griffin/NASA-HQ (Ed., 1/20/25). NOTE: Contract number changed to JPL 1699891 per J. Callas/JPL (Ed., 8/29/24). NOTE: End date changed to 9/27/2024 per U. Israelsson/JPL (Ed., 10/20/21) NOTE: End date changed to 3/31/2022 per B. Carpenter/NASA HQ (Ed., 1/4/2021) NOTE: New end date is 10/30/2020 per JPL (Ed., 5/21/19)		
Key Personnel Changes/Previous PI:	A new postdoc Dr. Abel Beregi was hired in January 2025, joining Dr Jean-Baptiste Gerent who was hired in April 2024.		
COI Name (Institution):	Aveline, David Ph.D. (Jet Propulsion Laboratory) Vishveshwara, Smitha Ph.D. (University of Illinois at Urbana-Champaign)		
Grant/Contract No.:	JPL 1699891 ; JPL 1502172		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Notions of geometry, topology, and dimensionality have directed the historical development of quantum-gas physics. With a toolbox of forces used to confine, guide, and excite Bose-Einstein condensates (BEC) or degenerate Fermi gases (DFG), physicists have used quantum gases to test fundamental ideas in quantum theory, statistical mechanics, and in recent years notions of strongly-correlated many-body physics from the condensed-matter world. We propose a specific program to explore a trapping geometry for quantum gases that is both tantalizing theoretically and prohibitively difficult to attain terrestrially: a quantum gas in a bubble geometry, i.e., a trap formed by a spherical or ellipsoidal shell structure, confining a 2D quantum gas to the surface of an experimentally-controlled topologically-connected “bubble.” The physics of a quantum gas confined to such a surface has not been explored terrestrially due to the limitations of gravitational sag; interesting work has certainly been done with gases confined to the lower regions of bubble potentials, but the fully symmetric situation has yet to be explored. The low-energy excitations of such a system are unexplored, and notions of vortex creation and behavior as well as Kosterlitz-Thouless physics are tantalizing aims as well. The solid-state modeling goals of the optical-lattice physics community are also fundamentally connected to the system, as the canonical Mott-insulator/superfluid transition features superfluid shells isolated between insulating regions. The central method to reach the sought-after bubble-geometry BEC or DFG is that of rf or microwave dressing of the bare trapping potentials provided by the Cold Atom Laboratory (CAL) “chip trap.” Radiofrequency dressing has been used conceptually through "rf-knife" evaporative cooling, but more recently through explicit construction of adiabatic potentials for interferometry, and shell-trap construction for both thermal and quantum gases. The proposed work is a window into a physical regime that is quite difficult to achieve terrestrially due to trap distortion; given the advantages of a microgravity environment, NASA CAL is uniquely positioned to realize the physics goals of this proposal.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work, while focused on the fundamental physics of ultracold atoms and not directly connected to human life, has a similar impact to life on Earth as that of all fundamental physics; it broadens our understanding of the physical world and helps us further cement our collective picture of quantum mechanics as "the way the world works." It explores the limits of how large Bose-Einstein condensates can be made, and to what extent the gravity-well of terrestrial labs render certain investigations difficult or impossible. The observations made aboard CAL through this project are a clear demonstration that physical insight can sometimes require microgravity facilities to be fully developed, and that spaceborne atomic physics experiments can be valuable contributions to our collective scientific efforts.
Task Progress:	The 2024-2025 period of this work was initially focused on planning and execution of ultracold atomic physics experiments aboard the Cold Atom Laboratory (CAL) with the "SM3B" generation of hardware that was launched in August 2023. This hardware experienced technical issues and was temporarily replaced with "SM1" hardware that produced ultracold samples through June-December 2024 and is ongoing. Initial studies focused on transport of ultracold samples away from the atom-chip surface and measuring residual 'sloshing' amplitudes; this was in preparation for creation of rf-dressed bubbles of greater homogeneity and improved aspect ratio compared with the work of previous years. Modeling efforts focused on understanding the magnetic-field variation in the SM1 hardware and the effects on ultracold bubble formation. We have found the relevant radiofrequency (rf) resonances of multiple CAL traps and characterized their mechanical oscillation frequencies. Following resonance-finding we were able to demonstrate (in August 2024) that rf-dressing for bubble generation was feasible on SM1 hardware. Current efforts will focus on demonstrating adiabatic transport and radiofrequency dressing into a aspect-ratio ~2 bubble (in contrast to aspect ratio 3 or 4 bubbles of previous work). We have returned to calculations and discussions with Barry Garraway of Sussex University regarding techniques of compensating rf inhomogeneity using microwave radiation, in a process known as 'dual dressing'. Such a technique might be possible aboard CAL. Dr. Lundblad was central planner of this work in this period, together with our partner at Jet Propulsion Laboratory (JPL), David Aveline, who, in addition to service as Co-Investigator, was our primary liaison to experimental operations aboard CAL. Postdoctoral scholars Gerent and Beregi at Bates College have supported the project through experimental planning and modeling. Theory Co-Investigators Smitha Vishveshwara and senior graduate student Brendan Rhyno provided regular insight and support, especially in regard to computational modeling of observed phenomena. Rhyno defended his PhD thesis in November 2024 (see bibliography), representing the culmination of many years of work investigating the physics of ultracold bubbles. The interesting phenomena that this project focuses on was dissemenated in many research-group seminars, department seminars, conference presentations, and posters.
Bibliography Type:	Description: (Last Updated: 07/13/2026)
Abstracts for Journals and Proceedings	Williams J, Aveline D, Botsi S, Elliott E, Kohel J, Kellogg J, Lay N, Sbroscia M, Schneider C, Thompson R, Bigelow N, Cornell EA, Engels PW, Lundblad N, Sackett CA, Oudrhiri K. "Space-enabled quantum science with NASA's Cold Atom Lab (CAL) operating on the International Space Station." 55th Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics, Fort Worth, TX, June 3-5, 2024. Bulletin of the American Physical Society. 2024 Jun 6:S00.00145. https://meetings.aps.org/Meeting/DAMOP24/Session/S00.145 , Jun-2024
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Dissertations and Theses	Rhyno B. "Explorations of engineered bosonic systems in ultracold atoms, oscillator networks, quantum spin chains, and space." Dissertation, University of Illinois, November 2024. , Nov-2024
Significant Media Coverage	Vishveshwara S, Vishveshwara CV. "Cold Atom Lab and Nathan Lundblad mentioned in popularizing-science history/memoir." Two revolutions: Einstein's relativity and quantum physics: A dialogue between father and daughter. Oxford Academic Press, Dec. 9, 2024. https://doi.org/10.1093/9780191818080.001.0001 , Dec-2024